Beyond Parallel Corpora

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data and machine learning

Supervised and Unsupervised



- We framed machine translation as a supervised machine learning task
  - training examples with labels
  - here: input sentences with translation
  - structured prediction: output has to be constructed in several steps

- Unsupervised learning
  - training examples without labels
  - here: just sentences in the input language
  - we will also look at using just sentences output language

- Semi-supervised learning
  - some labeled training data
  - some unlabeled training data (usually more)

- Self-training
  - make predictions on unlabeled training data
  - use predicted labeled as supervised translation data

Transfer Learning

- Learning from data similar to our task

- Other language pairs
  - first, train a model on different language pair
  - then, train on the targeted language pair
  - or: train jointly on both

- Multi-Task training
  - train on a related task first
  - e.g., part-of-speeh tagging

- Share some or all of the components

using monolingual data

Using Monolingual Data

- Language model
  - trained on large amounts of target language data
  - better fluency of output
- Key to success of statistical machine translation
- Neural machine translation
  - integrate neural language model into model
  - create artificial data with backtranslation

Adding a Language Model

- Train a separate language model
- Add as conditioning context to the decoder
- Recall state progression in the decoder
  - decoder state s_i
  - embedding of previous output word Ey_{i-1}
  - input context c_i

$$s_i = f(s_{i-1},\ Ey_{i-1}, c_i)$$

- Add hidden state of neural language model s_i^{LM}

$$s_i = f(s_{i-1},\ Ey_{i-1}, c_i, s_i^{\text{LM}})$$

- Pre-train language model
- Leave its parameters fixed during translation model training

Refinements

- Balance impact of language model vs. translation model
- Learn a scaling factor (gate)

$$\text{gate}_i^{\text{LM}} = f(s_i^{\text{LM}})$$

- Use it to scale values of language model state

$$\bar{s}_i^{\text{LM}} = \text{gate}_i^{\text{LM}} \times s_i^{\text{LM}}$$

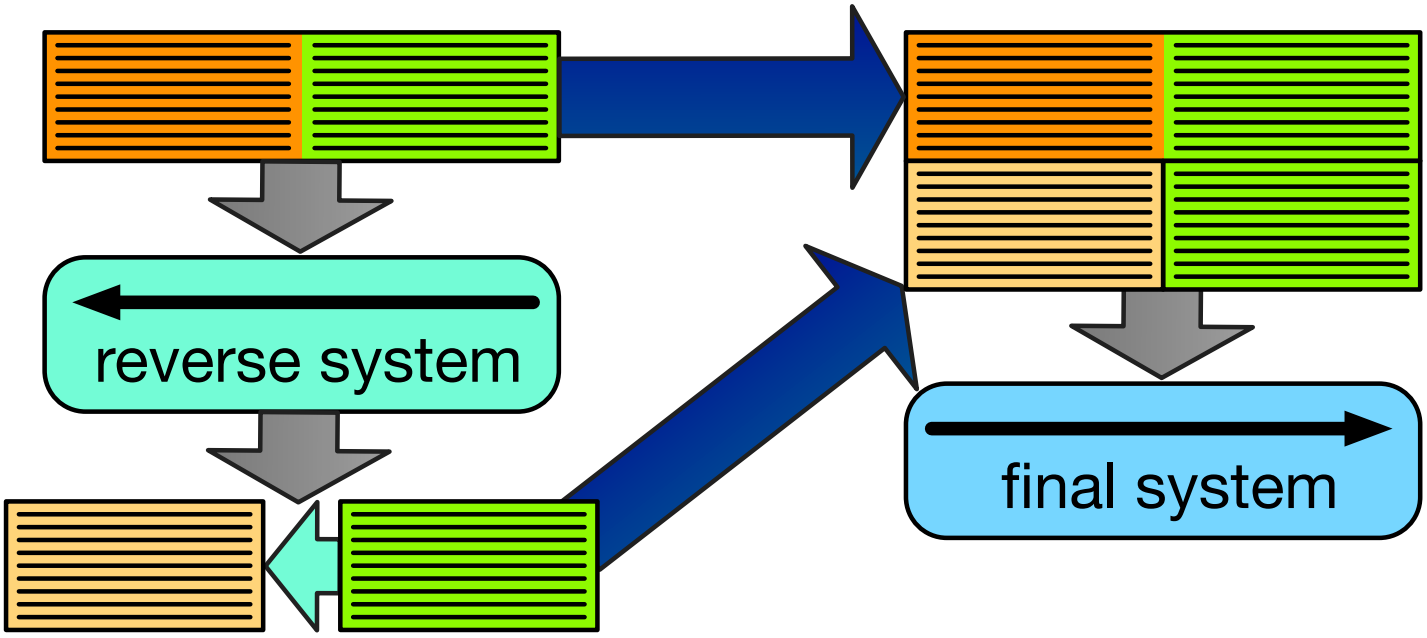
- Use this scaled language model state for decoder state

$$s_i = f(s_{i-1},\ Ey_{i-1}, c_i, \bar{s}_i^{\text{LM}})$$

backtranslation

Back Translation

- Monolingual data is parallel data that misses its other half
- Let's synthesize that half



Back Translation

- Steps

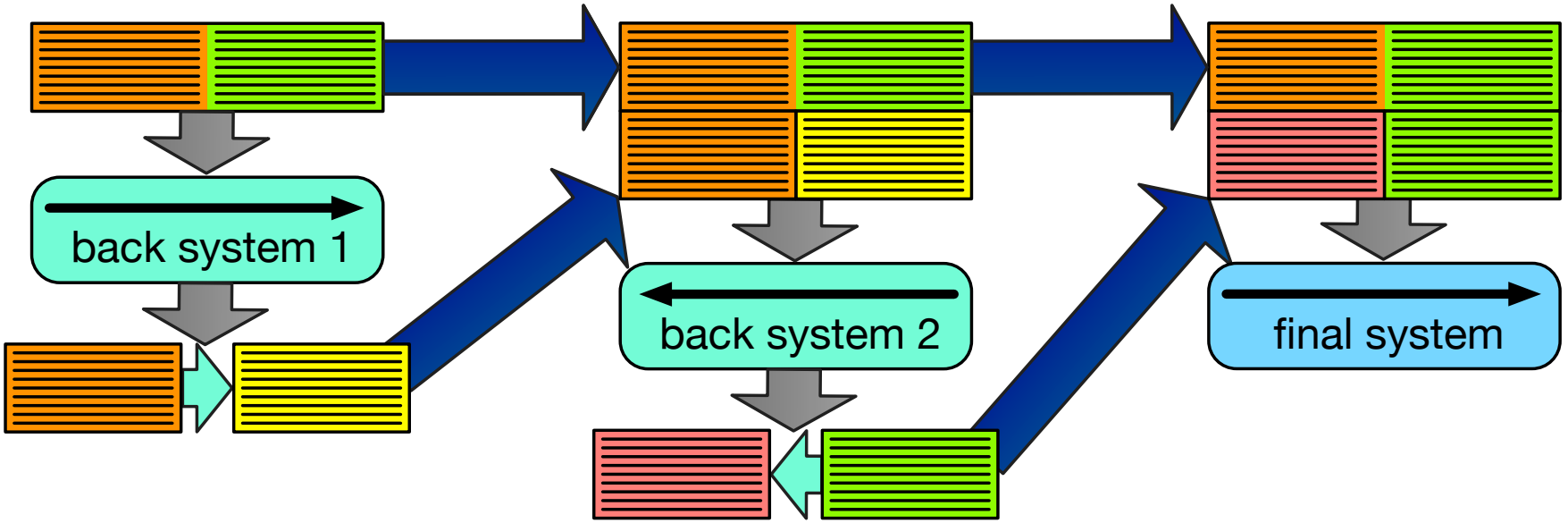
  1. train a system in reverse language translation
  2. use this system to translate target side monolingual data
     $\rightarrow$ synthetic parallel corpus
  3. combine generated synthetic parallel data with real parallel data to build the final system

- Roughly equal amounts of synthetic and real data

- Useful method for domain adaptation

Iterative Back Translation

- Quality of backtranslation system matters
- Build a better backtranslation system ... with backtranslation



Iterative Back Translation

- Example: Better system for backtranslation matters

German–English	Back	Final
no back-translation	-	29.6
*10k iterations	10.6	29.6 (+0.0)
*100k iterations	21.0	31.1 (+1.5)
convergence	23.7	32.5 (+2.9)
re-back-translation	27.9	33.6 (+4.0)

* = limited training of back-translation system

Variants

- Copy Target
  - if no good neural machine translation system to start with
  - just copy target language text to the source

- Forward Translation
  - synthesize training data in same direction as training
  - self-training (inferior but sometimes successful)

Round Trip Training

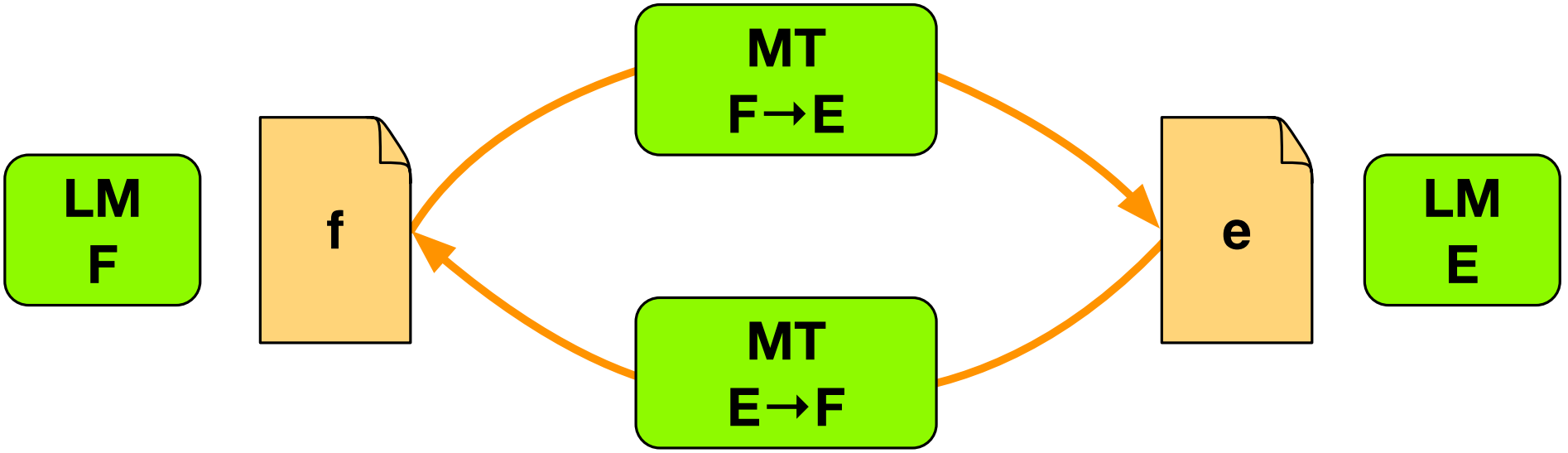
- We could iterate through steps of
  - train system
  - create synthetic corpus

- Dual learning: train models in both directions together
  - translation models $F \rightarrow E$ and $E \rightarrow F$
  - take sentence **f**
  - translate into sentence **e'**
  - translate that back into sentence **f'**
  - training objective: **f** should match **f'**

- Setup could be fooled by just copying (**e'** = **f**)

  $\Rightarrow$ score **e'** with a language for language E
  add language model score as cost to training objective

Round Trip Training



monolingual pretraining

Low Resource Language Pairs

- Problem: not enough parallel to even train a proper encoder or decoder
- Idea: use monolingual data
  - ... in source language → initialize encoder
  - ... in target language → initialize decoder
- How do we present monolingual data in training?

Masked Training

- Replace some input word sequences with <pad> (30% of words)
- Train model MASKED → TEXT on both source and target text

Why did the chicken cross the road?

⇑

Why did <pad> chicken <pad> the road?

Reordering Sentences

- Reorder sentences (each training example has 3 sentences)

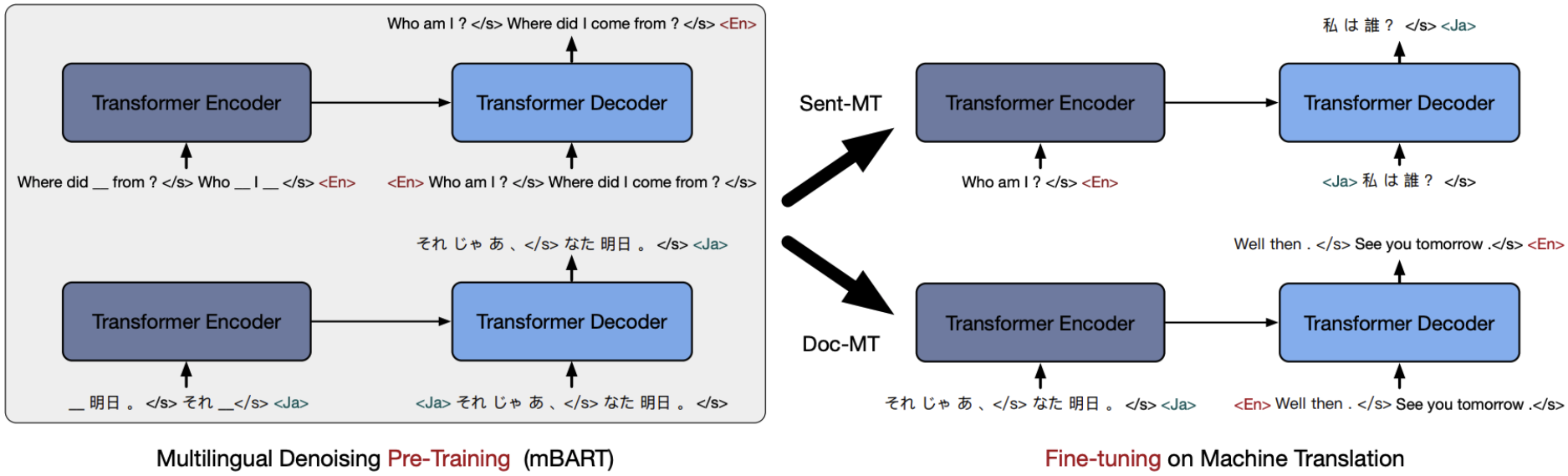
Why did the chicken cross the road?
The chicken wanted to get to the other side.
There are some delicious sunflower seeds.

⇓

The chicken wanted to get <pad> other <pad>.
<pad> are some delicious <pad> seeds.
Why did <pad> chicken <pad> the road?

Example: mBART

"Multilingual Denoising Pre-training for Neural Machine Translation" (Liu et al., 2020)

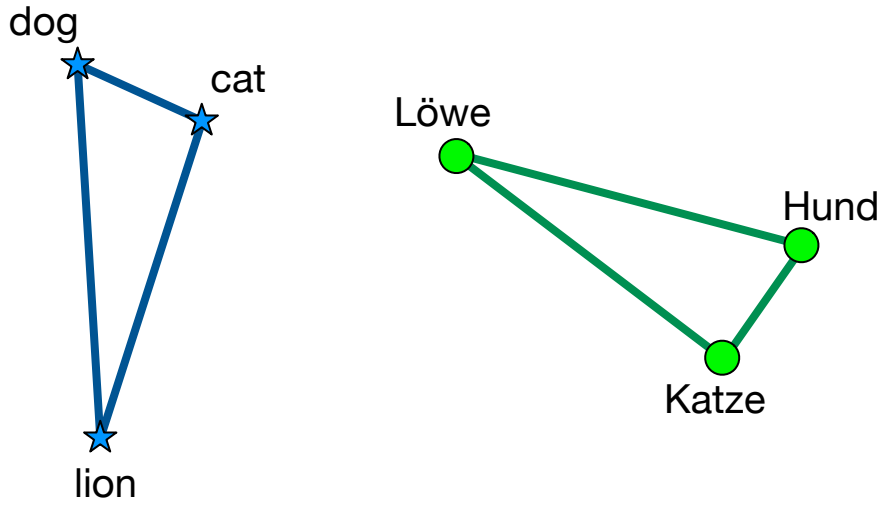


- 25 languages: from 55 billion words English to 56 million words Burmese
- Followed by training on parallel data

$\Rightarrow$ Helps with low-resource languages
(but not with >20 million sentence pair parallel data)

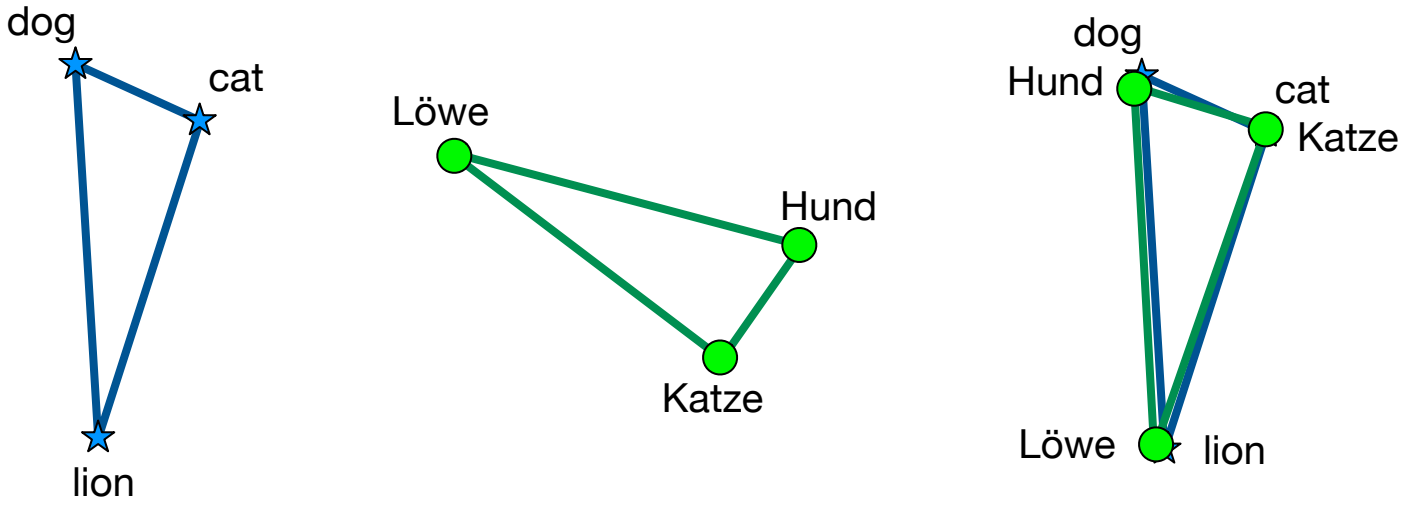
unsupervised machine translation

Monolingual Embedding Spaces



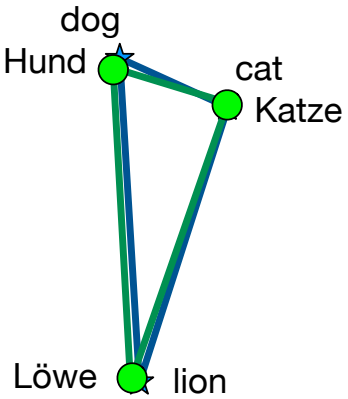
- Embedding spaces for different languages have similar shape
- Intuition: relationship between *dog*, *cat*, and *lion*, independent of language
- How can we rotate the triangle to match up?

Matching Embedding Spaces



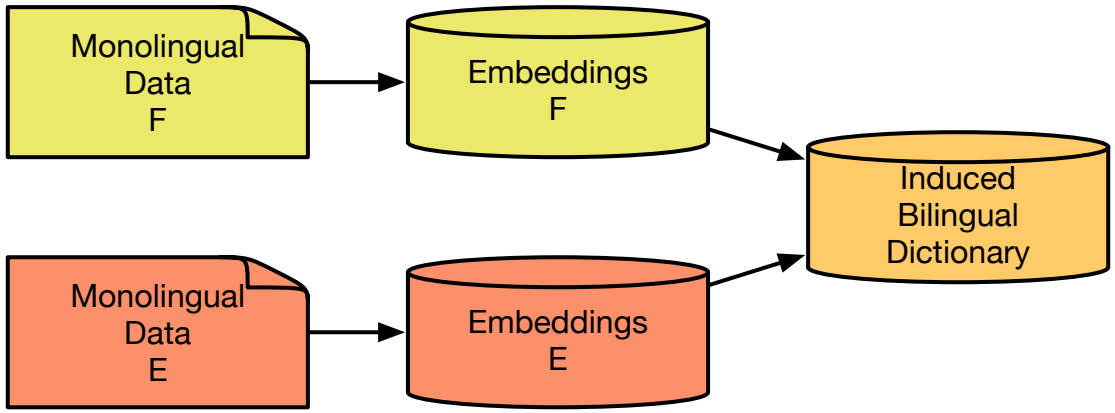
- Seed lexicon of identically spelled words, numbers, names
- Adversarial training: discriminator predicts language [Conneau et al., 2018]
- Match matrices with word similarity scores: Vecmap [Artetxe et al., 2018]

Bilingual Lexicon Induction



- Given shared embedding state

$\Rightarrow$ matching points in space = word translations



Inferred Translation Model

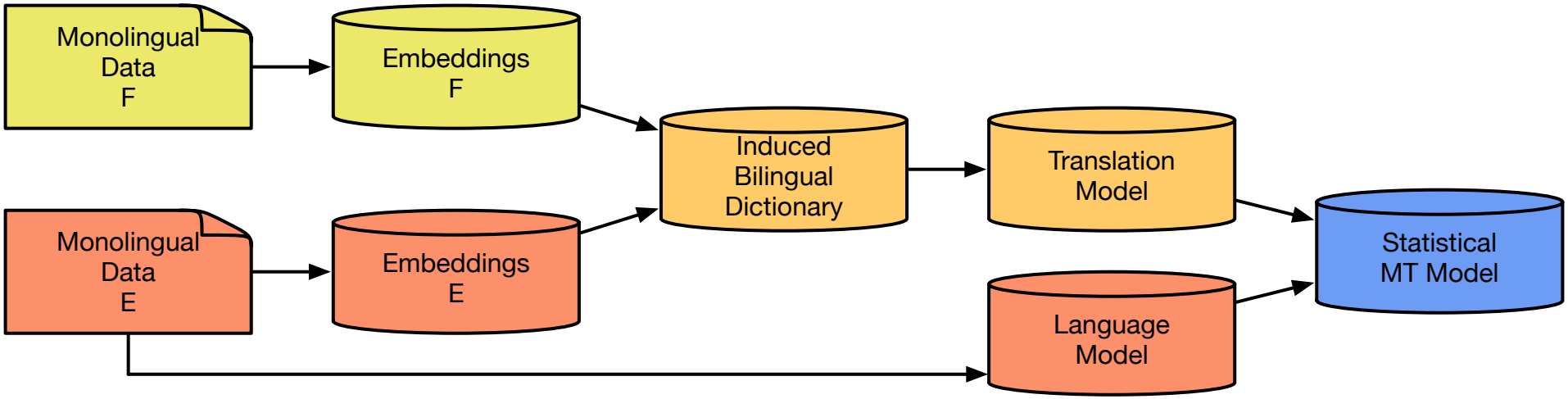
- Translation model
  - induced word translations

  → statistical phrase translation table (probability ≃ similarity)

- Language model
  - target side monolingual data

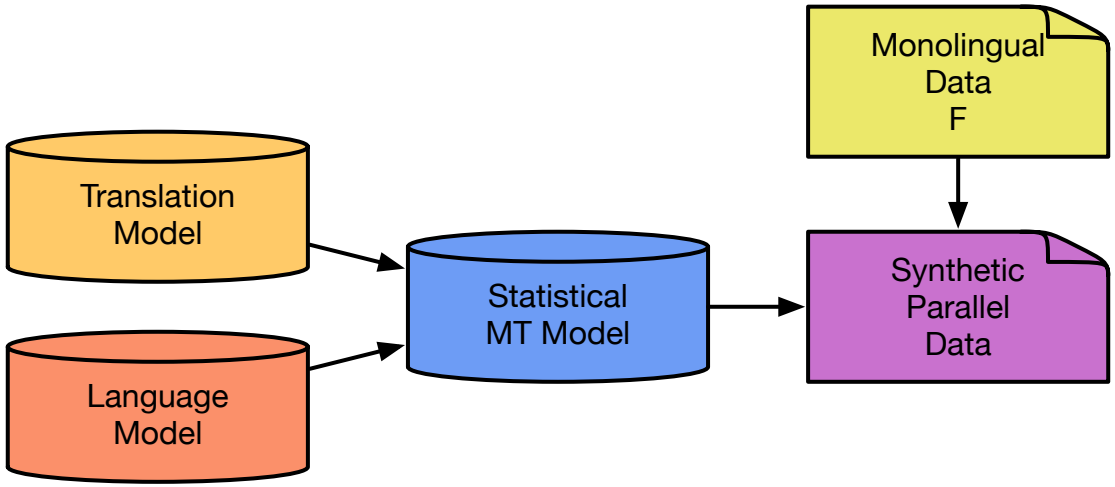
  → estimate statistical n-gram language model

⇒ Statistical phrase-based machine translation system



Synthetic Training Data

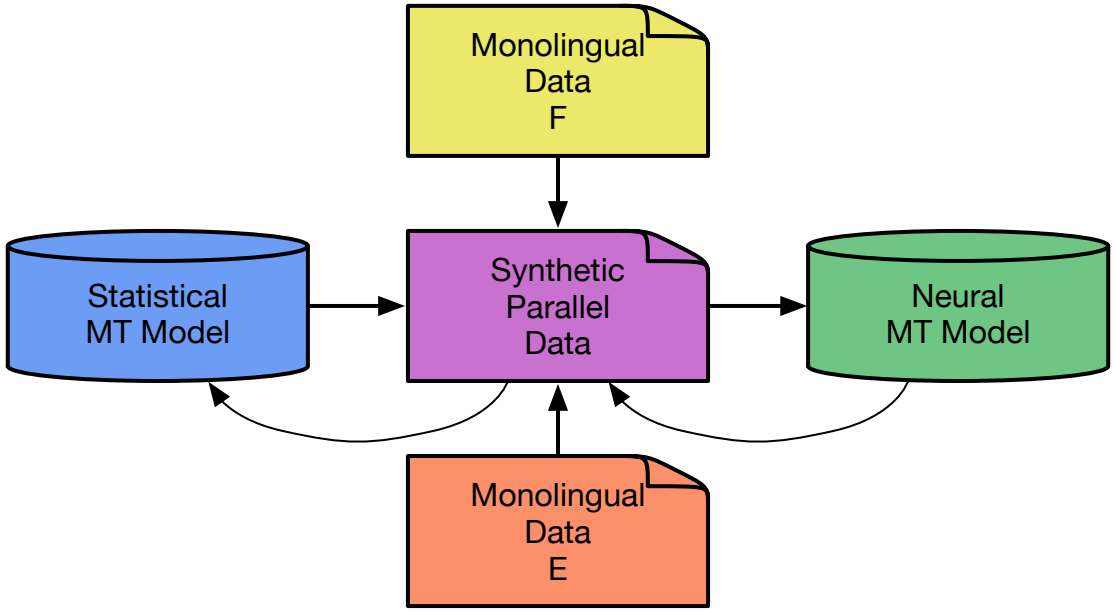
- Create synthetic parallel corpus
  - monolingual text in source language
  - translate with inferred system: translations in target language



Iterate

- Iterate
  - Predict data: generate translation for monolingual corpus
  - Predict model: estimate model from synthetic data
  - iterate this process, alternate between language directions

- Increasingly use neural machine translation model to synthesize data



multiple language pairs

Multiple Language Pairs

- There are more than two languages in the world
- We may want to build systems for many language pairs
- Typical: train separate models for each
- Joint training

Multiple Input Languages

- Example
  - German–English
  - French–English
- Concatenate training data
- Joint model benefits from exposure to more English data
- Shown beneficial in low resource conditions
- Do input languages have to be related? (maybe not)

Multiple Output Languages



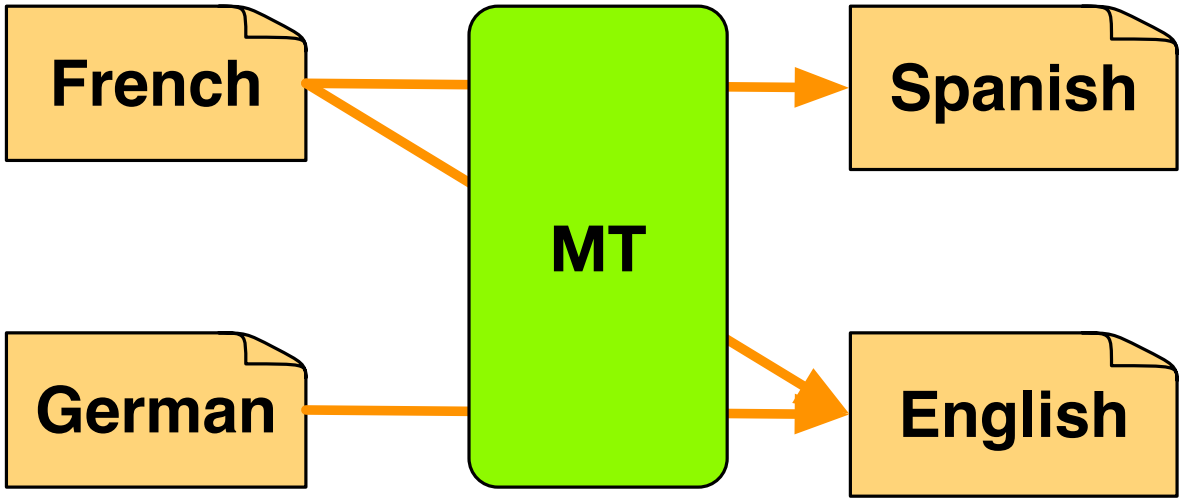
- Example
  - French–English
  - French–Spanish
- Concatenate training data
- Given a French input sentence, how specify output language?
- Indicate output language with special tag

[ENGLISH] *N'y a-t-il pas ici deux poids, deux mesures?*
⇒ *Is this not a case of double standards?*

[SPANISH] *N'y a-t-il pas ici deux poids, deux mesures?*
⇒ *¿No puede verse con toda claridad que estamos utilizando un doble rasero?*

Zero Shot Translation

- Example
  - German–English
  - French–English
  - French–Spanish
- We want to translate
  - German–Spanish



Zero Shot

- Train on
  - German–English
  - French–English
  - French–Spanish

- Specify translation

[SPANISH] *Messen wir hier nicht mit zweierlei Maß?*
⇒ *¿No puede verse con toda claridad que estamos utilizando un doble rasero?*

Zero Shot: Hype

Algorithms

Google's AI just created its own universal 'language'

The technology used in Google Translate can identify hidden material between languages to create what's known as interlingua

By **MATT BURGESS**

23 Nov 2016

Zero Shot: Reality

Table 5: Portuguese→Spanish BLEU scores using various models.

	Model	Zero-shot	BLEU
(a)	PBMT bridged	no	28.99
(b)	NMT bridged	no	30.91
(c)	NMT Pt→Es	no	31.50
(d)	Model 1 (Pt→En, En→Es)	yes	21.62
(e)	Model 2 (En↔{Es, Pt})	yes	24.75
(f)	Model 2 + incremental training	no	31.77

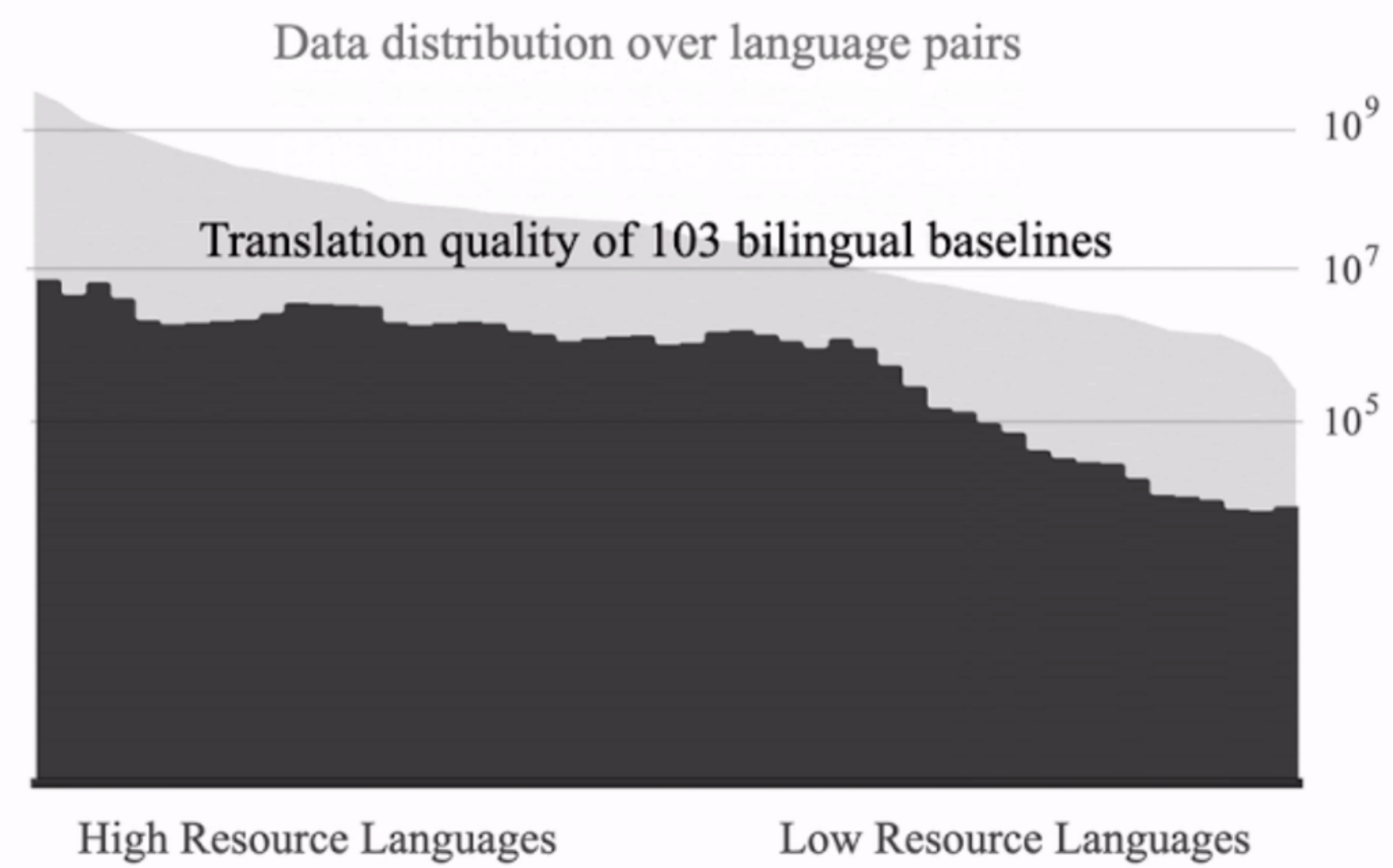
- Bridged: pivot translation Portuguese → English → Spanish
- Model 1 and 2: Zero shot training
- Model 2 + incremental training: use of some training data in language pair

Massively Multilingual Training



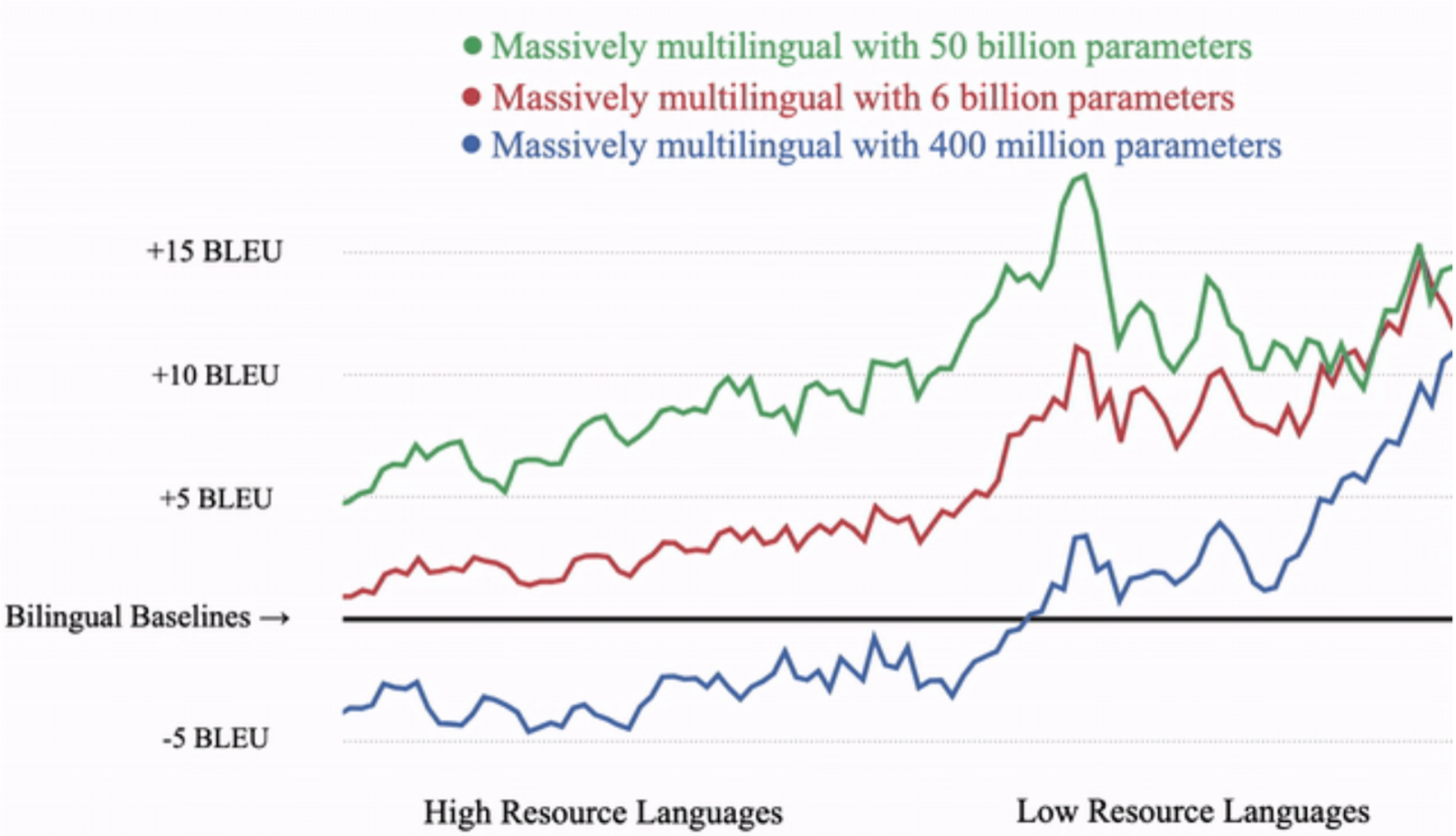
- Scaling up multilingual machine translation for more languages
  - many-to-English
  - English-to-many
  - many-to-many
- Mainly motivated by improving low-resource language pairs
- Move towards larger models

Translation Quality for 103 Languages



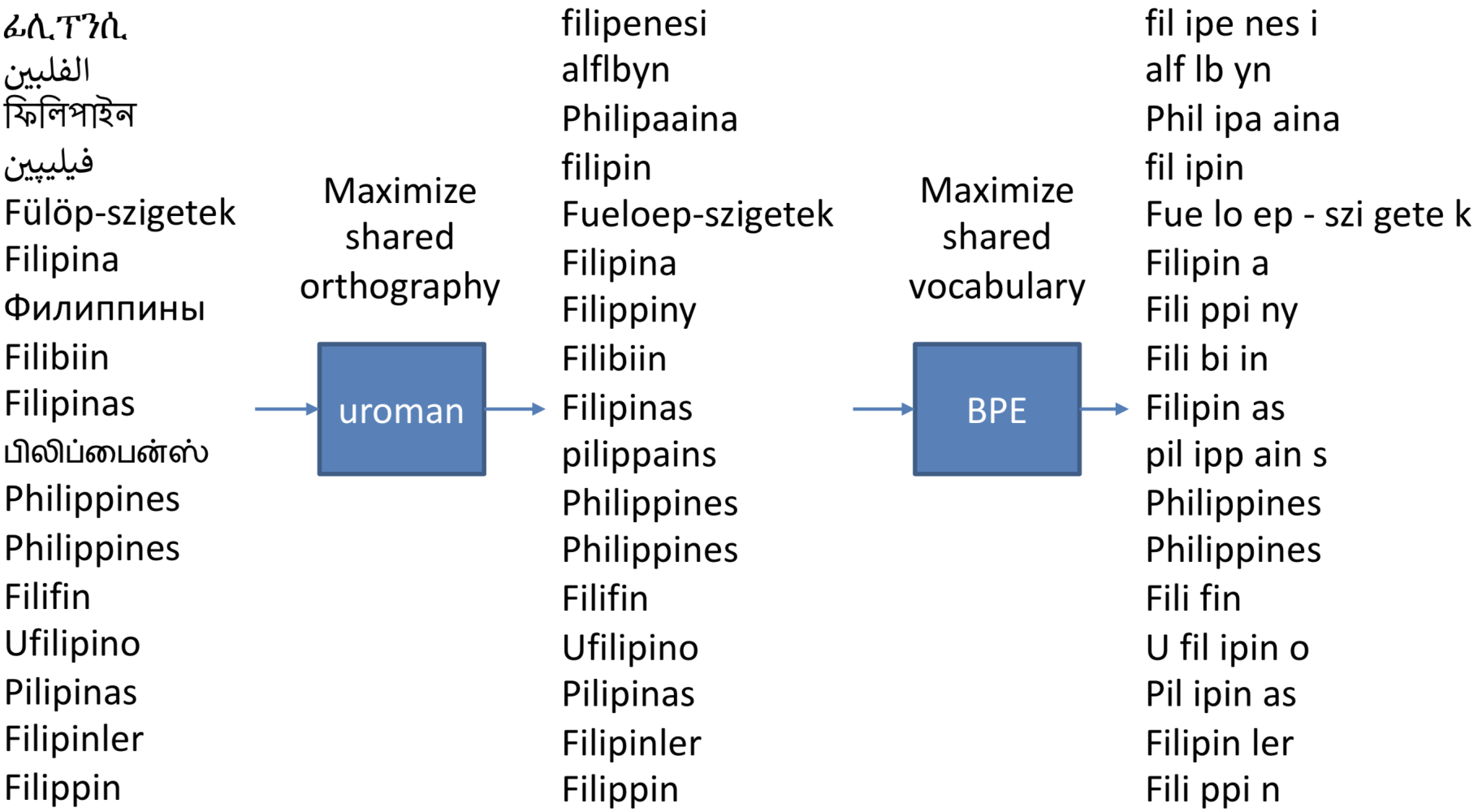
(source: Google)

Gains with Multilingual Training



(source: Google)

Romanization



(source: USC/ISI)

Many-to-Many

Facebook

Introducing the First AI Model That Translates 100 Languages Without Relying on English

October 19, 2020
By Angela Fan, Research Assistant

- 7.5 billion sentences for 100 languages
  (mined from web-crawled data)
- Model with 15 billion parameters
- Improvements especially for low resource languages

Even Bigger: NLLB (2022)



- No Language Left Behind: 200 languages
- Hand-translated test set: Flores-200
- Uses diverse data sources
  - public parallel data
  - translations created by professional translators
  - sentence pairs based on sentence embedding similarity
  - monolingual data for
    - monolingual pre-training
    - back-translation
    - self-training
- Models of different scale (up to 54B parameters), publicly released

Different Amounts of Data per Language



- High-resource language pairs are undertrained
- Low-resource language pairs are overtrained

$\Rightarrow$ Oversampling low resource language pairs
Data selection probability p_l for language pair l based on corpus sizes D_k

$$p_l = (D_l / \sum_k D_k)^{1/T}$$

- Curriculum training: adding low-resource data only in later training stages

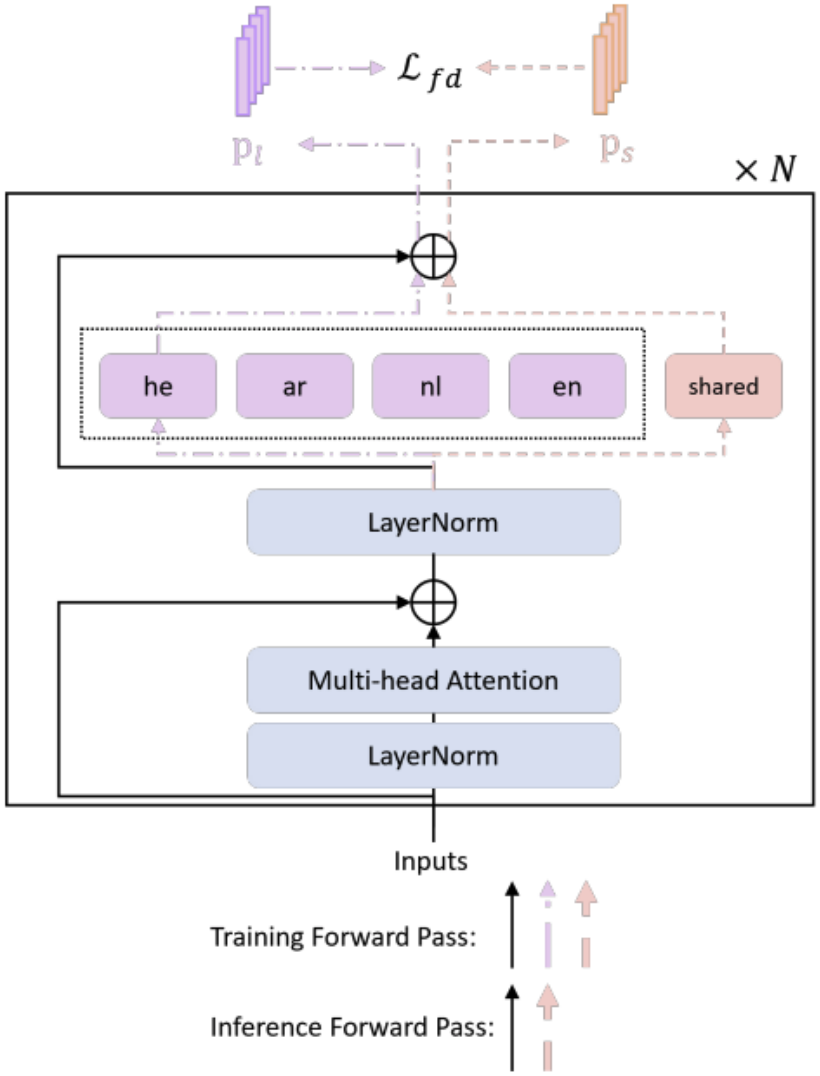
Interference

- Many languages in the same representation space
- Beneficial: shared cognates, numbers, names, ...
- Harmful: a lot of accidental overlap in tokens that have different meaning
  - *die* — common German determiner
  - *die* — different meaning in English
- What can be done to avoid harmful interference?

Language-Specific Components

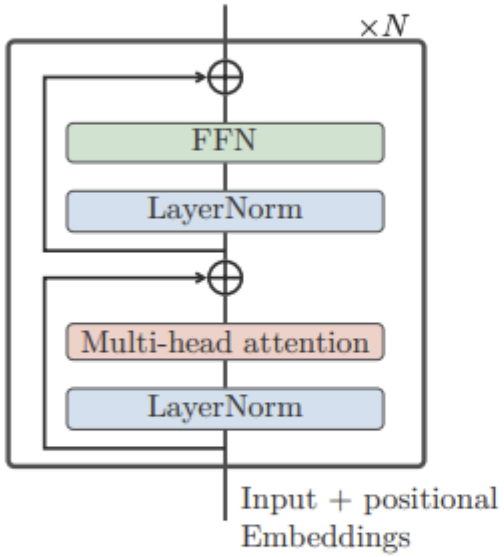
- Various design choices
  - language-specific encoder
  - language-specific decoder
  - language specific adaptor components

- Example:
  "Condensing Multilingual Knowledge with Lightweight Language-Specific Modules"
  Xu et al. (2023)
  - language specific parameters
  - shared parameters
  - self-distillation method to condense everything into shared parameters

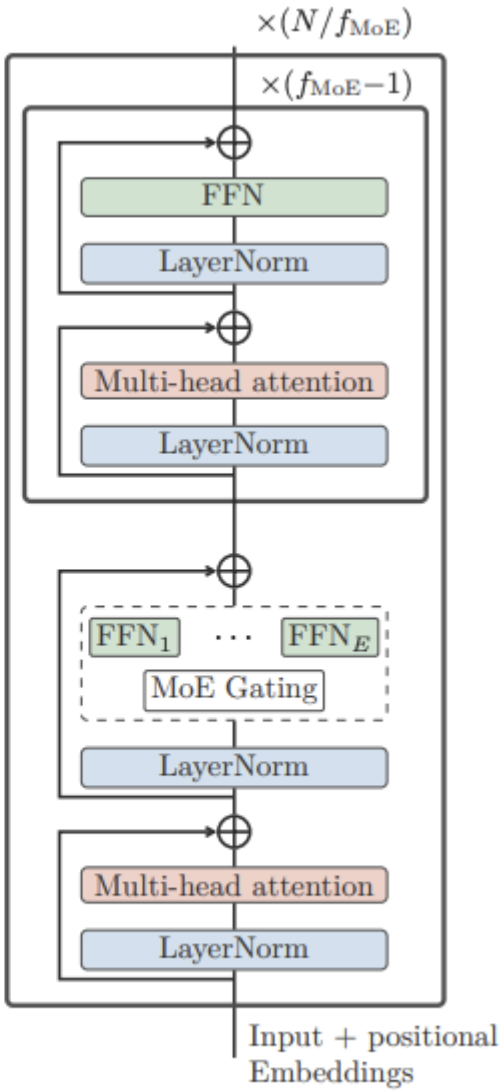


Mixture of Experts

- Conditional compute
- Gating mechanism decides which FF step to utilize
- Allows scaling to many more parameters without increasing computational cost



(a) Dense Transformer



(b) MoE Transformer

document-level translation

The Importance of Document-Level Context



- Pronouns
  - *I bought a table. It is pretty.*
  - *Ich kaufte einen Tisch. Er/sie/es is schön.*
- Better disambiguation
  - I have a lot of numbers. I still need to make the table.
- Terminological consistency

Why Not Document-Level Translation?



- Entire infrastructure focused on sentence level
  - Training data available as sentence pairs
  - Metrics defined at sentence level
  - APIs typically operate at sentence level

- This is slowly changing
  - Scaling up transformers for multi-sentence translation [Junczys-Dowmunt et al., 2019]
  - Document-level metrics, e.g., CTXPRO [Wicks et al., 2023]
  - Release of training data in document-aligned format
    e.g., Europarl, News Commentary, Paracrawl [Wicks et al., 2024]



questions?